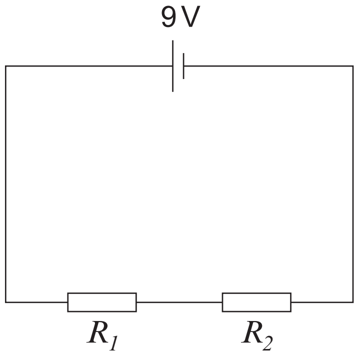
Examiner only

**5.** Jamie is investigating an electrical circuit which has two **identical** resistors, $R_1$ and $R_2$. The circuit that he built is shown below.




(a) Complete the sentences by underlining the correct word(s) in the brackets.

(i) In this series circuit the current in $R_1$ is **( the same as / higher than / lower than )** the current in $R_2$. [1]

(ii) In this series circuit the voltage across $R_1$ is **( the same as / higher than / lower than )** the voltage across $R_2$. [1]

(b) The voltage across $R_1$ is 4.5 V.

(i) Each resistor was found to have a resistance of 2.25 Ω.

I. Use the equation:

$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

to calculate the current in $R_1$. [2]

current = .......................................... A

II. Use the equation:

power = voltage × current

to calculate the power developed in resistor $R_1$. [2]

power developed in $R_1$ = .......................................... W



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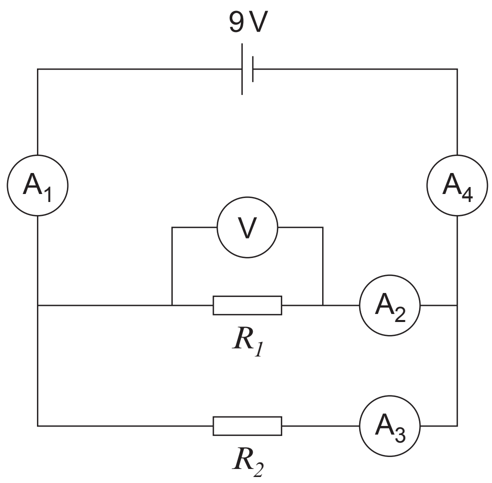
(ii) State the voltage across $R_2$. [1]

voltage across $R_2$ = .......................................... V

(c) Jamie then built a parallel circuit using the same 2.25 Ω resistors.




(i) State which ammeter (**$A_1$**, **$A_2$**, **$A_3$** or **$A_4$**) measures the current through $R_1$. [1]

......................................................................

(ii) State the voltmeter reading across $R_1$. [1]

voltmeter reading = .......................................... V

(iii) Ammeter $A_2$ reads 4 amps.

I. State the reading on ammeter $A_3$. [1]

ammeter reading = .......................................... A

II. Calculate the reading shown on ammeter $A_1$. [1]

ammeter reading = .......................................... A

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